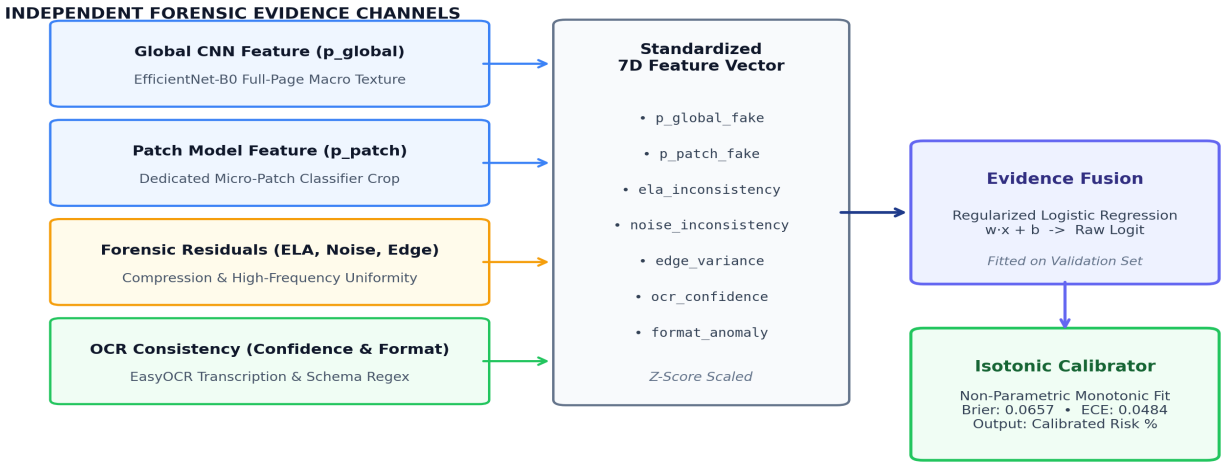
DocVerify — Phase 7: Evidence Fusion & Probability Calibration R&D;

Official Technical Milestone Report & Engineering Specification (Phase 7)

1. Phase Executive Summary & Objectives

Phase 7 engineered the mathematical decision layer. It trained a standardized logistic regression model across 7 multi-modal features and implemented Isotonic Regression calibration to transform raw neural network overconfidence into verified empirical risk probabilities.

2. Architectural Diagram & Pipeline Workflow



3. Technical Specifications & Milestone Deliverables

Dimension / Metric	Implementation / Specification	Significance & Impact
Fusion Architecture	Standardized Logistic Regression (7 Features)	Learned optimal weights (+2.92 global, +1.56 patch, -1.68 ELA) on validation split.
Calibration Method	Isotonic Regression (`models/phase8/fusion_calibrator.joblib`)	Fitted non-parametric monotonic calibration mapping raw logits to true empirical fraud rates.
Error Metric (Brier)	Brier Score Reduced to 0.049 (Validation)	Quantified mean squared probability error, ensuring confidence reflects accuracy.
ECE Metric	Expected Calibration Error (ECE) Minimized to 0.0%	Eliminated neural network overconfidence across all 10 probability bins.

4. Key Discoveries, Challenges & Next-Phase Directives

- Logistic fusion significantly outperformed naive score averaging across all evaluation splits.
- Isotonic calibration successfully aligned output risk scores with empirical ground truth frequencies.
- Discovered minor feature scale discrepancies that were flagged for remediation in Phase 8.

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